

Google OAuth2 Credential Setup for n8n (Gmail Integration)

A reference guide for connecting a self-hosted n8n instance to Gmail via Google OAuth2, based on a working setup.

Overview

- **n8n instance:** self-hosted (redirect URI uses `localhost`)
- **Redirect URI:** `http://localhost:5678/rest/oauth2-credential/callback`
- **Google Cloud project user type:** External
- **Publishing status:** Testing (recommended for personal/internal use — avoids full Google verification)

Step-by-Step Setup

1. Create/select a Google Cloud project

Go to [Google Cloud Console](#) and select or create the project you'll use for this integration.

2. Enable the Gmail API

[APIs & Services](#) → [Library](#) → search **Gmail API** → click **Enable**.

3. Configure the OAuth consent screen — Branding

[APIs & Services](#) → [OAuth consent screen](#) → [Branding](#)

Fill in:

Field	Value
App name	e.g. <code>n8n Gmail Integration</code>
User support email	your Gmail address
Developer contact email	your Gmail address

Leave blank (not required for Testing mode):

- App logo
- Application home page
- Application privacy policy link
- Application terms of service link
- Authorized domains

 **Avoid linking to `docs.google.com` or any Google-owned URL** as a privacy policy link — Google will then require you to authorize `google.com` as a domain, which you can't do (you don't own it). Just leave the field blank instead.

4. Set audience / user type

OAuth consent screen → Audience

- **User type:** External (Internal is only available for Google Workspace organizations, not personal Gmail accounts)
- **Publishing status:** Testing

5. Add yourself as a test user

Still on the **Audience** tab → **Test users** → **+ Add users** → enter your Gmail address → **Save**.

This is the step most likely to cause the **Error 403: access_denied** if skipped. A Testing-mode app with **zero test users** cannot be used by anyone, including you.

6. Configure scopes (Data Access)

OAuth consent screen → Data Access (or **Scopes**, depending on console version)

Add the scopes your n8n workflow needs. Common Gmail scopes:

- <https://www.googleapis.com/auth/gmail.labels>
- <https://www.googleapis.com/auth/gmail.compose>
- <https://www.googleapis.com/auth/gmail.modify>
- <https://mail.google.com/> (*full access — only include if you actually need it*)

💡 Narrower scopes are safer. If you don't need full mailbox access, skip mail.google.com and stick to [gmail.modify](https://www.googleapis.com/auth/gmail.modify) + [gmail.compose](https://www.googleapis.com/auth/gmail.compose) + [gmail.labels](https://www.googleapis.com/auth/gmail.labels).

7. Create OAuth 2.0 credentials

APIs & Services → Credentials → Create Credentials → OAuth client ID

- **Application type:** Web application
- **Authorized redirect URI:** <http://localhost:5678/rest/oauth2-credential/callback>

Copy the generated **Client ID** and **Client Secret**.

8. Configure the credential in n8n

In n8n, create a new **Gmail OAuth2 API** credential and paste in:

- Client ID
- Client Secret

Click **Connect my account**, complete the Google consent flow (you'll see an "unverified app" warning — click **Advanced** → **Go to [app name] (unsafe)** to proceed since it's your own app).

Troubleshooting

Symptom	Cause	Fix
Error 403: access_denied — app not verified	No test users added	Add your email under Audience → Test users
"OAuth configuration is incomplete" on Audience page	Missing required Branding fields, or zero test users	Fill required Branding fields (App name, support email, developer contact email); add a test user
"Invalid domain: must be a top private domain"	Privacy policy link points to a Google-owned domain (e.g. Google Docs)	Remove the privacy policy link — it's optional in Testing mode
Not billing/credit related	—	OAuth setup and Gmail API (within free quota) don't consume Cloud billing credits

Notes

- **100 test user cap** applies for the lifetime of the app while in Testing mode — plenty for personal/internal use.
- Full public verification is only needed if you plan to publish the app for use by people outside your test user list, especially with sensitive/restricted scopes like mail.google.com.